

Asad Ullah Aziz

Senior Full Stack Engineer

📍 Lahore, Pakistan

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Summary

Senior Full Stack Engineer with 5+ years of experience building responsive, fault-tolerant web applications, distributed systems, event-driven microservices, and AI-powered SaaS products. Hands-on across the stack: React, Next.js, and TanStack Query on the frontend; Node.js, NestJS, FastAPI, and Spring Boot on the backend. At Xenia, led AI product engineering that directly contributed to a **\$12M Series A** and enterprise deals with **Dave's Hot Chicken** and **McDonald's USA**. Experienced in production LLM integration, RAG pipelines, and multi-agent workflows using LangChain, LangGraph, PGVector, and OpenAI. Proven technical leader with experience in system architecture, Technical Design Documents, and mentoring engineering teams.

Experience

Xenia

Senior Software Engineer

Sep 2024 – Present
Chicago, Illinois, USA - Remote

- Led AI feature development (AI Template Digitizer, Xenia AI Copilot, and AI Summaries) that directly contributed to Xenia's **\$12M Series A** and enterprise onboarding of **McDonald's USA** and **Dave's Hot Chicken**.
- Built AI Template Digitizer (LangChain, GPT-4o, RAG, PGVector) to convert paper inspection sheets into digital templates; **cut churn by 99%, boosted signups by 20%, improved prompt accuracy by 60%, and reduced generation time by 30%** via Langfuse-driven model optimization.
- Built Xenia AI Copilot, a multi-agent system (LangChain, LangGraph, PGVector, ClickHouse, OpenAI API), giving operations managers visibility across all business locations; **key feature in closing McDonald's USA**.
- Built an MCP server for Xenia, enabling LLMs to create and assign tasks, manage projects, and generate templates through natural language; reduced platform complexity for operations teams.
- Re-architected a cron-based notification system into event-driven microservices, eliminating lock contention, duplicate delivery, and cascading provider failures; **scaled from 100K to 5M+ daily notifications** across email, SMS, WhatsApp, and push.
- Built a PDF generation service (Handlebars, Puppeteer, Ghostscript) with an AWS Lambda pipeline for image compression; **cut document build time from 30+ minutes to under 1 minute (98% faster)** for image-heavy files.
- Built a real-time analytics dashboard (CDC over ClickHouse) and Scheduled Reporting feature (Trigger.dev, PostgreSQL); directly contributed to **onboarding Dave's Hot Chicken** and enabled operations managers to track task completion, audit results, and compliance on configurable schedules.
- Integrated Stytych device fingerprinting to block bot attacks peaking at **1M fraudulent requests/day**; achieved **100% elimination of bot traffic** with zero degradation to legitimate users.
- Established AI-assisted development workflows by configuring Cursor, Claude Code and OpenCode and adding custom agents, AI coding rules, AI skills, and knowledge documentation to improve code generation accuracy and developer productivity.

WriteSea

Senior Software Engineer

Jan 2024 – Sep 2024
New York, New York, USA - Remote

- Integrated Stripe Connect with Express Accounts for career coach payouts; **generated \$35K in revenue and disbursed \$28K to partners** within the first month of launch.
- Built a resume and cover letter generation service (**Docker, FastAPI, LangChain, OpenAI API, Pinecone**) delivering personalized, job-tailored content to individual career seekers; **key feature in closing the University of Alabama deal**.
- Built a Next.js frontend with TanStack Query for server state management, improving page load performance and reducing data-fetching complexity.
- Designed MongoDB schema and built a RESTful API with NestJS and Mongoose; integrated Apex Charts for analytics visualizations in the admin dashboard.
- Served as technical lead for 5 engineers; owned architecture design, Technical Design Documents, PR reviews, onboarding flows, and cross-OS dev environment setup.

Venturennox

Backend Engineer

Jun 2022 – Jan 2024

Lahore, Punjab - Hybrid

- As founding engineer of TestFuse, migrated a FastAPI monolith to a 5-service event-driven microservices architecture following 12-Factor App methodology; used Apache Kafka for inter-service communication and saga pattern for distributed transaction coordination.
- Built Strandguard auth microservice (NestJS, PostgreSQL) with multi-tenant RBAC, dynamic roles, resource scoping, tenant isolation, Google OAuth, and SSO.
- Built an end-to-end coding assessment feature with Judge0 for sandboxed multi-language execution; supported public and hidden test cases with per-language file structures managed via dependency injection.
- Built an AWS Lambda pipeline (S3 pre-signed URL trigger) using MediaConvert and Transcribe for video interview transcription, enabling automated candidate response evaluation.
- Integrated Stripe subscription billing with webhook handling for free and paid tiers; validated full checkout and upgrade flows using Stripe Time Machine.
- Integrated Elasticsearch and Logstash to collect, index, and expose application logs via the TestFuse API, giving developers real-time visibility into integration issues.

Sentwitt

Full Stack Engineer

May 2021 – Jun 2022

Lahore, Punjab - Remote

- Built responsive React UIs with Redux state management; developed an Express.js microservice with a GraphQL API for JWT-based authentication, backed by MongoDB for large-scale data storage.
- Fine-tuned a RoBERTa model using Hugging Face Transformers for project-specific NLP text classification; deployed it as a scalable Python Django REST Framework microservice.
- Developed a Scrapy-based web scraper for keyword and hashtag-driven content extraction from social media platforms at scale.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, SQL, C, C++, PHP

Backend: Node.js, NestJS, Express.js, FastAPI, Django REST Framework, Spring Boot, Laravel, REST API, GraphQL,

AI / LLM: LangChain, LangGraph, LangFuse, OpenAI API (GPT-4o), RAG, PGVector, Pinecone, Prompt Engineering, Hugging Face Transformers

Frontend: React, Next.js, Redux, TanStack Query, Tailwind CSS, Chakra UI, Apex Charts, Story Book

Databases: PostgreSQL, MongoDB, ClickHouse, Redis, Elasticsearch

Cloud & Infrastructure: AWS (ECS, EC2, Lambda, S3, MediaConvert, Transcribe), GCP, Docker, Apache Kafka, RabbitMQ, BullMQ, Trigger.dev, Ingest

Tools: Git, Bash, Stripe, Stytych, Puppeteer, Ghostscript, Judge0, Logstash, Jest, Sentry

Development: Claude Code, Cursor, OpenCode, Pi Agent

Education

The University Of Lahore

Bachelor of Science in Computer Science (GPA: 3.28 / 4.00)

2018 - 2022

Lahore, Punjab

- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Artificial Intelligence, Distributed Systems